

Day 112: Weekly Revision — Supervised ML and Evaluation

1. Day number

Day 112

2. Topic name

Revision of Days 106–111: Supervised Machine Learning and Evaluation

3. Connection

Over the last six days, you learned several beginner **supervised machine-learning models** and the metrics used to evaluate them.

The big picture is:

```
Labeled data
  ↓
Choose classification or regression
  ↓
Train a suitable model
  ↓
Make predictions
  ↓
Evaluate those predictions
```

Today is about connecting those pieces rather than learning a new algorithm.

4. Revision summary of Days 106–111

Day 106 — Logistic Regression

Logistic Regression is commonly used for **classification**, especially binary classification.

Example:

```
Customer features
  ↓
Logistic Regression
  ↓
Churn / No Churn
```

It can estimate a probability and then convert that into a predicted class.

Remember:

| Despite its name, Logistic Regression is primarily a classification algorithm.

Day 107 — K-Nearest Neighbors

KNN classifies a new example by looking at its nearby training examples.

```
New point
  ↓
Find k nearest points
  ↓
Look at their classes
  ↓
Majority vote
  ↓
Predicted class
```

Main idea:

| Nearby examples help decide the new example's class.

Day 108 — Naive Bayes

Naive Bayes uses **probability**.

It asks:

| Given the evidence I see, which class appears more likely?

For spam classification:

```
Words/features
  ↓
Probability evidence
  ↓
Spam vs Not Spam
```

The "naive" part comes from its simplifying conditional-independence assumption about features.

Day 109 — Decision Tree

A Decision Tree predicts by following simple learned decisions.

Think of your early Python if-else statements:

```
Question?
  ↓
Yes / No
  ↓
Another question?
  ↓
Leaf prediction
```

Important tree terms:

Root
Split
Branch
Leaf

Day 109 — Random Forest

Random Forest extends the Decision Tree idea.

Instead of using one tree:

Tree → prediction

it uses many:

Tree 1 ┘
Tree 2 ┘
Tree 3 ┘ → combine predictions → final result
Tree 4 ┘
Tree 5 ┘

For classification, the trees can effectively vote.

Day 110 — Classification evaluation

For classification, you learned:

True Positive
True Negative
False Positive
False Negative

These form the **confusion matrix**.

From them, you can calculate:

Accuracy
Precision
Recall
F1 score

Each metric answers a slightly different question.

Day 111 — Regression evaluation

For numerical predictions, you learned:

MAE
MSE
RMSE

All measure prediction error differently.

You also learned **cross-validation**, where different portions of the available training data take turns being used for validation.

5. Important topics recap

Topic	Beginner meaning
Logistic Regression	Probability-based binary classifier
KNN	Classify using nearby examples
Naive Bayes	Classify using probability and evidence
Decision Tree	Follow learned yes/no decisions
Random Forest	Combine many Decision Trees
Confusion matrix	Count types of classification results
Accuracy	Overall fraction classified correctly
Precision	How reliable positive predictions are
Recall	How many actual positives were found
F1	Balance of precision and recall
MAE	Average absolute regression error
MSE	Average squared regression error
RMSE	Square-root version of MSE
Cross-validation	Evaluate repeatedly using different folds

6. Foundational notes

Before choosing an algorithm or metric, ask the most important question:

| **What type of target am I predicting?**

If the target is a category:

Yes / No
Spam / Not Spam
A / B

you have a:

Classification problem

If the target is a continuous number:

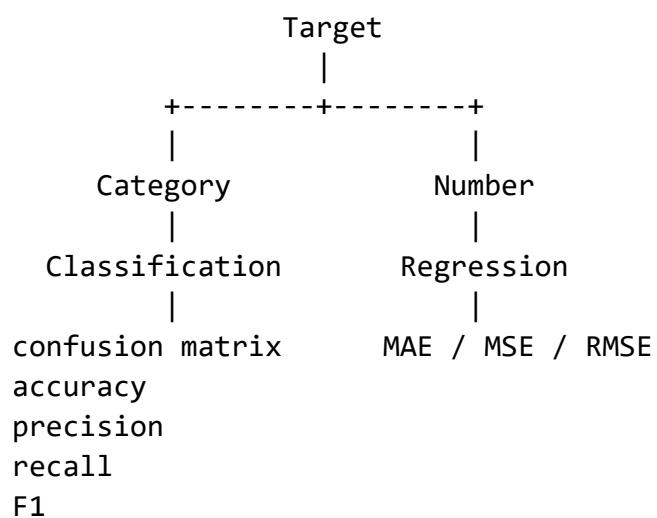
House price
Temperature
Sales amount

you have a:

Regression problem

This immediately affects both your model and your evaluation metrics.

A simple mental map is:



7. Easy example

Imagine a real-estate dataset containing:

house_size
bedrooms
location_score

We could create two different ML problems from similar features.

Problem A

Predict:

Will the house sell within 30 days?

Possible target:

Yes / No

This is:

Classification

Possible models:

Logistic Regression
KNN
Naive Bayes
Decision Tree
Random Forest

Possible evaluation:

Confusion matrix
Accuracy
Precision
Recall
F1

Problem B

Predict:

What will the house sell for?

Possible target:

₹62 lakh
₹78 lakh
₹1.1 crore

This is:

Regression

A beginner model could be Linear Regression from Day 104.

Possible evaluation:

MAE
MSE
RMSE

Cross-validation can also help estimate how consistently the model performs across different subsets of the available training data.

8. Revision problem statement

You have two small ML problems.

Problem A — Classification

A company has:

monthly_usage
support_calls

and wants to predict:

churn = Yes / No

Your job:

1. Identify whether this is classification or regression.
 2. Choose one reasonable beginner classification model.
 3. Identify useful classification metrics.
 4. Explain what a false negative would mean if **Churn = Positive**.
-

Problem B — Regression

You have:

house_size
bedrooms

and want to predict:

house_price

Your job:

1. Identify whether this is classification or regression.
2. Choose an appropriate model type.
3. Identify suitable regression metrics.
4. Explain what MAE would mean.

5. Explain conceptually how cross-validation could evaluate the model several times.

Do not build large models. Focus on choosing the correct **problem type, model family, and evaluation method**.

9. Concepts used

This revision combines three major stages.

Stage 1 — Define the problem

Identify:

```
Features → X
Target   → y
```

Then decide:

```
Classification?
or
Regression?
```

Stage 2 — Choose a model

For classification, beginner options now include:

```
Logistic Regression
KNN
Naive Bayes
Decision Tree
Random Forest
```

For simple numerical prediction, you already know:

```
Linear Regression
```

Stage 3 — Evaluate

Classification:

```
Confusion matrix
Accuracy
Precision
Recall
F1
```

Regression:

MAE
MSE
RMSE

Cross-validation:

Repeat evaluation across different folds

10. Thought process

When someone gives you an ML problem, don't start by immediately choosing an algorithm.

Use this sequence.

First ask:

What is my target?

If it is:

0 / 1
Yes / No
A / B

think:

Classification

If it is:

52.4
₹750000
28.7°C

think:

Regression

Then choose a sensible beginner model.

After training, ask:

How should I measure mistakes?

For classification:

What classes were predicted correctly?
What kinds of classification errors occurred?

For regression:

How far are predictions from actual numbers?

Finally, consider whether a single train/validation split gives enough information or whether cross-validation would provide a broader view.

11. Beginner-friendly pseudocode

START

identify features X

identify target y

IF y contains categories:

 problem = classification

 choose a classifier

 train classifier

 make predictions

 build confusion matrix

 calculate useful metrics:

 accuracy

 precision

 recall

 F1

ELSE IF y contains continuous numerical values:

 problem = regression

 choose regression model

 train model

 make numerical predictions

 calculate:

 MAE

 MSE

 RMSE

OPTIONALLY:

 divide training data into folds

 repeat training and validation
 with different folds

 summarize cross-validation results

END

12. Suggested solving approach — conceptual model selection

Use a simple three-question method.

Question 1: What am I predicting?

Category → Classification
Number → Regression

Question 2: What model makes sense?

For a beginner classification exercise:

Logistic Regression → useful starting classifier
KNN → classify using nearby points
Naive Bayes → probability/evidence approach
Decision Tree → learned if-else rules
Random Forest → many trees combined

You do **not** need to decide that one of these is universally "best." Different algorithms behave differently depending on the dataset.

For a simple regression exercise:

Linear Regression

is a good starting point.

Question 3: What should I evaluate?

Classification
↓
Confusion matrix
Accuracy
Precision
Recall
F1

Regression
↓
MAE
MSE
RMSE

Then consider cross-validation when you want evaluation across multiple data splits.

13. Easy edge cases

Imbalanced classification data

Suppose:

95 customers → No Churn 5 customers → Churn
--

A model predicting No Churn every time gets:

95% accuracy

but detects no churn customers.

Therefore:

| Accuracy alone can be misleading.

Look at precision, recall, F1, and the confusion matrix.

No positive predictions

If a classifier never predicts the positive class, normal precision calculation becomes problematic because there are no positive predictions to evaluate.

This is another reason to inspect model behavior rather than trusting one metric blindly.

One huge regression error

Suppose errors are:

1 2 2 30

That 30 will strongly influence:

MSE RMSE

because the errors are squared.

MAE is generally less strongly affected by that single large error.

Perfect predictions

Classification:

FP = 0
FN = 0

Regression:

all errors = 0

In the regression case:

MAE = 0
MSE = 0
RMSE = 0

Tiny dataset

With very little data:

- classification metrics may vary greatly,
- regression errors may be unstable,
- individual examples can strongly affect the model,
- cross-validation folds can become extremely small.

Do not assume that a complicated model or many folds can replace having useful data.

14. Common mistakes to avoid

1. Using a regression model when the target is a category.

First identify the target type.

2. Using classification metrics for numerical predictions.

Accuracy does not tell you how close two house-price numbers are.

3. Looking only at accuracy.

Especially avoid this with imbalanced classes.

4. Confusing precision and recall.

Remember:

Precision:

"When I predicted positive, how often was I right?"

Recall:

"Of all actual positives, how many did I find?"

5. Thinking lower MAE/MSE/RMSE is worse.

These measure errors, so generally:

smaller error → better

6. **Assuming Random Forest is simply one large tree.**
It is a collection of multiple trees.
 7. **Thinking cross-validation means training one model once.**
The train/validation roles rotate across folds.
 8. **Choosing an algorithm before understanding the target.**
Start from the problem, not the model name.
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15. Quick self-check questions

1. You want to predict whether a transaction is Fraud or Not Fraud. Is this classification or regression?
2. Which metric asks, "Of all actual positive examples, how many did my classifier find?"
3. What is the basic difference between a Decision Tree and a Random Forest?
4. You predict house prices. Which group of metrics is appropriate: accuracy/F1 or MAE/MSE/RMSE?
5. Why might cross-validation give you more information than evaluating using only one validation split?

Try answering these without looking back.

16. Hint only

For the revision problems, begin with the **target**, not the algorithm:

```
Churn = Yes / No
      ↓
  ?

House price = numerical amount
      ↓
  ?
```

Then connect each problem to its metric family:

```
Classification
→ confusion matrix
→ accuracy / precision / recall / F1

Regression
→ prediction errors
→ MAE / MSE / RMSE
```

For cross-validation, remember:

Fold A validates once
Fold B validates once
Fold C validates once
...

The central idea from **Day 112** is:

First identify whether the task is classification or regression, then choose a suitable supervised model, and finally evaluate it with metrics that match the type of prediction.